

Activity 2.3.2 Ethics in AI

PLTW

ETHICS IN AI



An Ethical Future



Ethics in Action



Conclusion

ACTIVITY SUPPORTS



Source

An Ethical Future

GOALS	RESOURCES
• Investigate trade-offs in artificial intelligence. • Conduct research to determine potential solutions to future artificial intelligence problems. • Create and deliver an effective presentation.	
GOALS	RESOURCES
• <u>Professional Ethics</u> • <u>Principles of Ethics</u> • <u>Effective Research</u> • <u>Information Credibility</u> • <u>Citing Sources</u>	

- **Activity 2.3.2 Ethics in AI (Downloadable PDF)**

In Activity 2.3.1 AI Robotic Greetings, you were introduced to some of the capabilities of artificial intelligence (AI). As you can imagine, the capabilities of this technology are expanding rapidly and being implemented throughout society even faster.

AI is used in a variety of applications, like smart cars and smart homes, to make life easier. It is also used to help target advertisements that are shown to you while you browse the web.



However, with the implementation of this new technology, many trade-offs need to be considered. For example, software errors will occur in smart cars and people may get hurt, and information you thought was private may be collected without your knowledge. Many ethical issues arise with this new technology, and important decisions must be made by engineers.

In this activity, you will investigate two AI scenarios and determine the best ethical solutions.

Code of Ethics

In addition to undergraduate and graduate engineering degrees, some engineers decide to obtain a credential called a Professional Engineer (PE) license. To maintain the license, professional engineers are required to abide by a specific code of ethics. Two essential principles state: “Hold paramount the safety, health, and welfare of the public” and “Perform services only in areas of their competence.” For more examples, review the [Professional Ethics](#) resource.

Turn and Talk



What are some other ethical guidelines that you think engineers should follow?

Why do you think this?

Ethics in Action

Some ethical problems are simple, such as not cheating on a test, even if the teacher isn't looking, or not taking something that isn't yours, even if you won't get caught. However, as technology gets more complex, so does its ethical concerns. To solve complex ethical problems, engineers need to develop well-thought-out solutions.

1

Review the [Principles of Ethics](#) resource.

2

Form a team, review the following scenarios, and brainstorm possible solutions for two of them.



Ethical Scenario

- Transportation of goods across the country accounts for roughly two million jobs in the United States, specifically trucking. With the implementation of autonomous trucks, projections are that more than 90% of those jobs could be replaced by AI, leaving millions unemployed. What should be done about this issue?
- [Deepfake](#) technology is improving rapidly, which poses considerable ethical concerns. Should this technology be regulated? If so, how?
- AI has been and will continue to be used throughout the web, particularly on social media sites, to manipulate users through subtle deception to benefit advertisers, political leaders, and many others. How should the collection of data to manipulate behavior be addressed?

- When robots, like robot vacuums and smart home tools, are introduced into homes, they will be able to collect data on every aspect of people's homes and anything said within them. What should be done to protect individuals' rights to privacy in their homes?
- Jobs in fields like industry, transportation, military, research, and retail could be completed by robots in the future. With a potential to put millions of people out of work, what should our society do to help those who are affected by this technology?

3

Conduct research on the two scenarios your team has chosen and develop detailed solutions to each problem. Review the [Effective Research](#), [Information Credibility](#) and [Citing Sources](#) resources as needed.

HELPFUL HINT: Each scenario has multiple problems that need to be addressed. Recall and use the process you went through in Activity 0.1.2 What's the Problem to identify multiple problems for a situation.

4

Present your solutions to your class.



Real-World Robotics: Smart Tissue Autonomous Robot (STAR)

Surgery has come a long way throughout the course of modern medicine. Incisions are much smaller than in decades past, resulting in fewer complications and shorter recovery times. However, smaller incisions place greater demands on surgeons, who must precisely cut and suture despite limited maneuverability and obstructed views. To counteract that, robotic-assisted surgery places

the surgeon behind a computer console to control a robot that is holding the surgical implements. The next logical step: autonomous surgery!

Engineers at Johns Hopkins University in Baltimore, Maryland have developed the Smart Tissue Autonomous Robot (STAR) that can plan, adapt, and execute a laparoscopic surgical procedure in soft tissue with minimal human intervention. So far the robot has only practiced on the soft tissue of a pig, but the robot has demonstrated greater accuracy and consistency than expert surgeons' hands. This brings modern medicine one step closer to fully automated surgery.

IMPORTANT: The Johns Hopkins University STAR video contains graphic imagery of robotic surgery on a pig's intestinal tract. Viewer discretion is advised.



Johns Hopkins University STAR video¹

Conclusion

Question 1

After researching different AI problems, what rules or guidelines do you think can be applied across all of AI to prevent additional ethical concerns?

Question 2

Which presented solutions were the most interesting to you? Do you think they can be implemented? Why or why not?

Source

Third-party images, videos, and Historical Highlight information used throughout this activity were drawn from the following source:

1. Saeidi H, Opfermann JD, Kam M, Wei S, Leonard S, Hsieh MH, Kang JU, Krieger A. Autonomous Robotic Laparoscopic Surgery for Intestinal Anastomosis. Science Robotics. 7, no. 62 (2022). PMID: 35080901. DOI: 10.1126/scirobotics.abj2908